ELG 3125 - Signals & Systems

Lecture 9 - Part 1

Fourier Series Representation of Periodic Signals

Signal can be represented using **complex exponentials** - continuous-time and discrete-time Fourier series and transform.

If input to LTI system is expressed as **a linear combination of periodic complex exponentials** or sinusoids, the output can also be expressed in this form.

The Response of a LTI system to Complex Exponentials

Continuous time: $e^{st} \rightarrow H(s)e^{st}$

Discrete time: $z^n \rightarrow H(z)z^n$

s, z are complex variables

- An output which is a possibly complex number times the input is called an eigenfunction of the system.
- The Amplitude factor is referred to as the system's eigenvalue.

$y(t) = \lambda x(t)$ — $y(t)$: eigenfunction; λ: possibly complex; $|\lambda|$: amplitude.

input $x(t)$ applied to LTI system with impulse response of $h(t)$

$\Rightarrow$ output is: $y(t) = \int_{-\infty}^{\infty} h(\tau)\, x(t-\tau)\, d\tau$

let $x(t) = e^{st} \Rightarrow x(t-\tau) = e^{s(t-\tau)}$

$$\Rightarrow y(t) = \int_{-\infty}^{\infty} h(\tau) e^{s(t-\tau)} d\tau = \int_{-\infty}^{\infty} h(\tau) e^{st} e^{-s\tau} d\tau = e^{st} \underbrace{\int_{-\infty}^{\infty} h(\tau) e^{-s\tau} d\tau}_{= H(s)}$$

assume that it converges

$\Rightarrow H(s) = \int_{-\infty}^{\infty} h(\tau) e^{-s\tau} d\tau$

$\Rightarrow \boxed{y(t) = H(s)e^{st}}$

↳ because $x(t) = e^{st}$, complex exponentials e^{st} are eigenfunctions of LTI systems

↳ for specific values of $s \Rightarrow H(s)$ would be eigenvalues.

Looking at discrete-time LTI systems

$x[n] = z^n$

$y[n] = \sum_{k=-\infty}^{\infty} h[k]\, x[n-k]$, $x[n-k] = z^{n-k}$

$$\Rightarrow y[n] = \sum_{k=-\infty}^{\infty} h[k] z^{n-k} = \sum_{k=-\infty}^{\infty} h[k] z^n z^{-k} = z^n \underbrace{\sum_{k=-\infty}^{\infty} h[k] z^{-k}}_{H(z) = \sum_{k=-\infty}^{\infty} h[k] z^{-k}} = z^n H(z)$$

$\Rightarrow \boxed{y[n] = H(z) z^n}$

$H(z)$: eigenvalues; z^n: eigenfunctions

Benefits of decomposing general signals in terms of eigenfunctions for LTI system analysis

say: $x(t) = a_1 e^{s_1 t} + a_2 e^{s_2 t} + a_3 e^{s_3 t}$

$a_1 H(s_1) e^{s_1 t}$ $\quad a_2 H_2(s_2) e^{s_2 t}$ $\quad a_3 H_3(s_3) e^{s_3 t}$

$\Rightarrow y(t) = a_1 H_1(s_1) e^{s_1 t} + a_2 H_2(s_2) e^{s_2 t} + a_3 H_3(s_3) e^{s_3 t}$

Generalizing $\Rightarrow$ $x(t) = \sum_k a_k e^{s_k t}$ $\Rightarrow$ $\boxed{y(t) = \sum_k a_k H(s_k) e^{s_k t}}$

For discrete-time LTI system: if $x[n] = \sum_k a_k z_k^n$ $\Rightarrow$ $\boxed{y[n] = \sum_k a_k H(z_k) z_k^n}$